

C K M C A P I T A L

CKM Capital

Multi-Agent Macro Hedge Fund

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LangGraph DAG · 9 stages · ~30 instruments · ~\$1M paper NAV

What we built

A hypothesis-first macro fund, run by agents

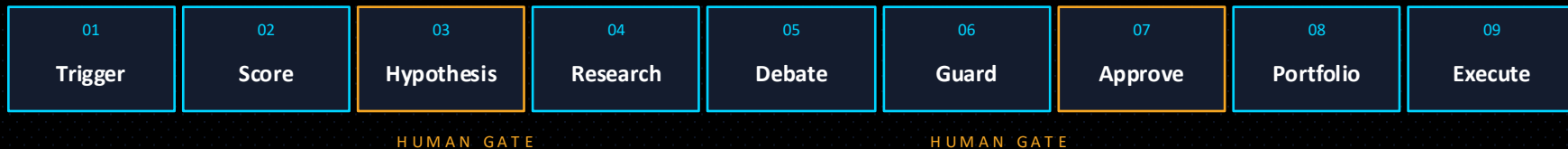
LLMs reason. Python enforces. Humans approve.

Multi-asset	Equities · Bonds · Commodities · FX · BTC (via IBIT)
Hypothesis funnel	Thesis → asset selection → research → debate → guard → portfolio
Constitution	Hard-coded rules + deterministic accounting floor
Status today	~30 instruments · ~\$1M paper NAV · 14 agents · 6 trigger paths · 2 HITL gates
Multi Modal	Text, news, PDFs, live policy speech (Fed / ECB via Deepgram STT), X tweets — multi-lexicon, persona-relative deviation scoring

From news to position

9 stages, 12+ LLM calls, 2 human gates

≈ \$0.30 per pipeline run · 6 trigger paths feed the pipeline (accumulated conviction, black swan, regime shift, calendar, figure deviation, cron)



Two-pass scoring

Bare LLM score → enriched re-score for borderlines (0.4–0.8) with Polymarket + X sentiment

Schema-enforced falsifiability

LLM cannot emit a thesis without a kill criterion

Sealed evidence

4 research agents compute in Python; LLM only interprets — cannot lie about RSI(14)

Adversarial debate

Bull vs Bear, adjudicated by a third agent citing the decisive argument

3 layers of risk

Constitutional Guard (deterministic) → Risk-Off Gate (XGBoost) → human

What feeds the pipeline?

Six data layers - all live - mostly free APIs

MACRO

FRED

CPI · NFP · PMI · yield curve · credit spreads · daily refresh

PRICES

yfinance · OpenBB

ETFs · futures · FX · 15-minute cache

NEWS

RSS + Sonar

Reuters · Fed · ECB · 200-word deep-read summaries

CALENDAR

FRED + Sonar

US releases via FRED, ECB / BoJ / OPEC via Sonar

ALT — MARKETS

Polymarket CLOB

Prediction-market prices on borderline headlines

ALT — SOCIAL

X / Twitter + Sonar

X API v2 tweet polling (Trump, Fed officials) + Sonar grounded sentiment search

Trump & Powell · figure-deviation pipeline

Persona-relative deviation, not absolute tone

Two figures · one normalized stream · z-scored against each speaker's own 12-month baseline

POWELL

Live policy speech → STT

Fed / ECB pressers transcribed sentence-by-sentence via Deepgram. Each sentence scored on hawkish · dovish · uncertainty lexicons.

TRUMP

X tweets → 5-min poll

X API v2 polled every 5 minutes. Each tweet scored in parallel on trade-protectionism · Fed-pressure · regulatory-stance lexicons.

Both streams → uniform FigurePost stream · downstream agents don't care where the signal came from

STAGE A · DETERMINISTIC

z-score > 1.5 σ

Each new sentence/tweet scored against the speaker's own 12-month rolling rhetorical baseline. A dovish Powell turning hawkish breaks >1.5 σ on his curve, not a generic hawkish threshold.

STAGE B · LLM + RAG

profile-card confirm

LLM pulls the figure's profile card — track record, behavioural tells, follow-through % — and confirms a genuine policy/stance shift, filtering quotes, hypotheticals, and asides.

BOTH PASS → FIRE

TriggerRecord(FigureDeviation)

Directional tags emitted — Powell → [rates_up, gold_dn], Trump → [usd_up, em_dn, semis] — then the standard agent pipeline runs unchanged.

One of six trigger paths — absolute tone is noise floor; deviation is the signal.

Critical fixes

Three architectural flaws, fixed

From 13 words of input to ~4,000 words of grounded context per run

Sparse input	Sonar deep-reads enrich significant headlines with ~200-word summaries; non-US calendar is live, not stub.
Single-event triggers	Directional Conviction Ledger accumulates 0.4–0.7 signals on a 12h half-life — captures 'five headlines, one regime' patterns.
Hindsight contamination	Replay script removed entirely. System runs forward-only on live data.
LLM mis-calibration	Two-pass scoring: bare LLM score, then re-score with Polymarket prices + X sentiment as evidence anchors.
Pure event reaction	Technical readiness multiplier (RSI / MACD / OBV) measures whether the market is primed before sizing the trade.

LangGraph DAG

9 stages - sealed boundaries

FundState flows append-only through every node



- **One LLM, one job** Tight Pydantic schema per agent, structured output enforced.
- **Hybrid agents** Research desk computes in Python; LLM cannot lie about RSI(14) — it copies verbatim.
- **Asymmetric memory** Every agent declares a `WriterIdentity`; out-of-matrix writes raise `WriteForbidden`.
- **Defense in depth** Portfolio Agent independently re-checks Guard veto before issuing an order.

Simulated advisor desk

Six economists & macro investors · three consultation modes

Advisors stress-test theses from named perspectives · not in the trade pipeline

THE ROSTER

Six macro personas

Krugman (Keynesian) · El-Erian (EM / cross-asset) · Summers (inflation hawk) · Druckenmiller (top-down PM) · Dalio (debt-cycle) · Tudor Jones (risk-off bias). Each backed by per-persona Chroma corpus + auto-generated belief card.

CONSULTATION MODES

Three modes + debate rooms

Per-turn chat · Parallel panel (PanelSynthesis: consensus, disagreements, strongest objection) · Multi-persona debate rooms (NDJSON streaming, round-robin sequential visibility — Krugman sees Summers' reply before answering, producing genuine debate dynamics).

- **Per-turn chat** Query embedded → top-k chunks → PersonaResponse with cited sources
- **Panel mode** asyncio.gather → PanelSynthesis · 'Apply Synthesis' copies result into approval notes
- **Standalone chat** 30-day sliding window · older messages visible but not billed · persisted to disk
- **Full audit trail** Consultation history attached to every ApprovalItem · preserved with each decision

Approval gates

Two checkpoints - full audit trail

The pipeline stalls until a human approves

GATE 01

Post-hypothesis

Full thesis text, kill criteria, regime, conviction surface in the dashboard before research begins.

GATE 02

Post-debate

Verdict with bull/bear reasoning surfaces before execution.

- **Approval queue** Disk-persisted JSON; cards show full payload context
- **Reasoning notes required** Auditable record of why each trade was killed
- **Persona advisor consultation** Chat · parallel panel with synthesis · multi-persona debate rooms at every gate
- **No auto-timeout** `wait_for_resolution()` polls until a human acts

Memory architecture + Curator

Short-term + long-term · asymmetric R/W · weekly distillation

Read everywhere · write narrowly

`short_term_journal.md`

Append-only log of every TriggerRecord, WorldStateBrief, Hypothesis, TradeExpression, ResearchBundle, Bull/Bear/Verdict, Guard decision, fill.

`long_term_journal.md`

Distilled lessons. Written only by Curator.

`conviction_ledger.json`

Decaying directional signal sums. Append-only by trigger runner.

Curator · weekly distillation

≥ 3 observations required per lesson · categories: regime_pattern, vehicle_preference, recurring_bias · lessons feed back to Hypothesis Agent · WriterIdentity enforces who writes what

The constitution

Human edited markdown - machine enforced

5 hard rules (deterministic) + 6 soft rules (LLM-evaluated)

HARD RULES

Python check · no override

- H1** 5% NAV / position · 40% / asset class · 10% drawdown freeze
- H2** $ADV \geq \$50M$ · position $\leq 1\%$ of ADV
- H3** VIX > 40 caps gross at 50% · 5d P&L < -5% freezes new positions
- H4** Only MARKET_OPEN, MARKET_CLOSE, TRIM, STOP_LOSS allowed
- H5** Every order must declare parent_hypothesis_id

SOFT RULES

LLM-evaluated · amendable

- S1** No averaging into a broken thesis
- S2** ≤ 2 trades per regime
- S3** 24h cooling-off after losses
- S4** Must cite kill criterion
- S5** 3-loss category review
- S6** Acknowledge contradicting LT lessons

Key insight: the LLM never sees hard rules. Python decides. LLM is short-circuited.

Frontend

Custom React - real-time polling

Nine routes · monochrome aesthetic · single-port deployment

React 18	TypeScript	Vite	Tailwind	Recharts	TanStack Query
Portfolio KPIs · positions · equity curve		Macro Regime quadrant · conviction ledger · risk-off gauge		Research Per-thesis evidence bundles	
Risk Exposure donut · warnings		Agents Per-agent feed · 14 agents		Approvals Pending hypotheses & verdicts · persona consult view	
Personas + Rooms Roster · standalone chat · multi-persona debate rooms		Approval Consult 3-col: item summary · persona picker + panel · apply-synthesis			

Polling: approvals 5s · portfolio 30s · charts 60s · macro 5m — pauses when tab hidden.

The Roster

Persona advisors · live view

Six personas · bios + flagship quotes · chat / expand entry points to consultation modes

RAG-backed replies

Per-persona Chroma

Belief cards

Persona advisors

Synthesize economists' and market investors. Consult them during HITL approval gates to test them against varied viewpoints, RAG-backed by their own writings — every reply cites the source. It draws from:

Personas are advisors only — they do not participate in the agent pipeline. To consult one, [go to Approvals](#) and click **Consult** — on a pending item.

PK **Paul Krugman**
Keynesian economist, NYT columnist

[Chat](#) [Expand](#)

Paul Krugman believes in the importance of evidence-based policy making and critiques ideological commitments that ignore empirical evidence, especially those detrimental to public welfare. He consistently highlights the negative impact of policies driven by greed and ignorance, particularly under administrations he perceives as violating these principles. Krugman is also an advocate for reforms addressing economic inequality and is critical of wealth displays that exacerbate social divisions. His economic analyses often focus on the implications of policy decisions on inflation, growth, and global financial positions.

Wild! Ignorance isn't neutral—it's deadly.

The books aren't cooked, they're just served with a side of ego.

Selling a \$500 million yacht is a rounding error for Bezos, but a symbol for the rest of us.

Leaving the world's rich means higher borrowing costs and weaker inflation down the line.

This isn't 1970s stagflation, but the risks are real.

HA **Mohamed A. El-Erian**
Chief economist, EY, former Citigroup advisor

[Chat](#) [Expand](#)

Mohamed A. El-Erian is a cautious observer of macroeconomic policies, with a keen focus on central bank decisions and their implications for financial stability. He emphasizes the interconnected nature of global economic risks and often questions the prevailing market sentiments to highlight potential vulnerabilities.

Defender of the Fed | Questioning conventional wisdom

Interconnected vulnerabilities

LS **Lawrence H. Summers**
Former Treasury Secretary, Harvard economist

[Chat](#) [Expand](#)

SD **Stanley Druckenmiller**
Investment manager, ex-Deutsche Capital

[Chat](#) [Expand](#)

RD **Ray Dalio**
Founder, Bridgewater Associates

[Chat](#) [Expand](#)

PJ **Paul Tudor Jones**
Hedge trader, founder of Tudor Investment Corp

[Chat](#) [Expand](#)

Advisors only

Personas never sit in the trade pipeline. Humans pull them in at HITL gates to stress-test theses.

Card anatomy

Each card: bio, signature quotes pulled from the corpus, Chat for per-turn dialogue, Expand for full profile.

Three modes

Per-turn chat · parallel panel with synthesis · multi-persona debate rooms with round-robin visibility.

Pipeline Inspector

Agent Pipeline · live view

186 entries · filter by stage · every hypothesis carries tags, kill criteria, full reasoning trace

Agent Pipeline Total Entries 186 Hypotheses 13 Verdicts 29 Guard Decisions 9 Lessons 0

All Stages Input Thesis Research Debate Control Memory

Trigger 3 Current Event 14 Hypothesis 13 Asset Selection 29 Research Desk 29 Bull 29 Bear 29 Debate 29 Guard 9 Curator 0

Hypotheses 13

2026-05-10 00:00

risk_off medium 21d 1 kill criteria

Over the next three weeks, Powell's Jackson Hole tighter-for-longer message should keep US policy expectations restrictive: front-end Treasury yields remain biased higher, the 2s10s curve stays under inversion pressure, the USD is supported, and risk assets face valuation headwinds as financial conditions tighten.

Macro regime context is not loaded, so regime framing must rely on the headline and leading indicator reads. The key leading indicators triggered are real_policy_rate, financial_conditions, and yield_curve_10y_2y. Powell's tighter-for-longer signal and the market repricing to two additional 25bp hikes imply a more restrictive real policy stance if inflation does not fall faster, tighter financial conditions through higher front-end rates/discount rates, and pressure for the 2y sector to remain elevated relative to the long end. That points to a near-term risk-off/tighter policy regime rather than the disinflation/easing regime reflected in some recent hypotheses. Conviction is medium, not high, because the signal is directly relevant and already visible in market pricing, but it is based on a single communication event and is vulnerable to reversal on soft inflation or

2026-05-09 00:00

risk_off medium 21d 2 kill criteria

Over the next three weeks, the Jackson Hole hawkish repricing should sustain a tighter-for-longer US policy regime: front-end yields remain biased higher, the USD stays supported, and equities face downside pressure as markets incorporate a more restrictive real policy-rate path.

Powell's Jackson Hole remarks were interpreted as more restrictive, and markets reportedly repriced to two additional 25bp hikes. The key leading indicator read is the real_policy_rate signal: a tighter-for-longer stance implies a more restrictive expected real policy-rate path, which should tighten financial conditions. With no loaded macro regime context, I frame this as a headline-driven restrictive-policy shock rather than a broad growth/inflation regime call. The closest top-down implication is higher front-end yields and firmer USD, with equities vulnerable as discount rates rise. Conviction is only medium because the evidence is a single policy-guidance surprise, the move is already partly priced, and prior disinflation/pause theses remain a credible counterweight if incoming inflation or labor data soften.

2026-05-04 00:00

disinflation medium 30d 2 kill criteria

US disinflation and a Fed pause regime persist over the next 30 days, limiting renewed tightening expectations and supporting easier financial conditions.

Core PCE at 2.4% YoY alongside explicit Fed pause signaling is a coherent top-down macro impulse: inflation pressure is moderating enough for the central bank to stop tightening. Over a 30-day horizon, that should keep the US in a disinflationary pause regime, reducing front-end rate pressure and generally easing financial conditions. Conviction is only medium because the brief contains one dominant data point and no corroborating labor, wage, ISM, or market-pricing evidence.

2023-11-03 15:42

risk_off medium 30d 1 kill criteria

The recent tariff imposition by Trump will induce a protective risk-off regime over the next 30 days, with continued downward

186

TOTAL ENTRIES

13

HYPOTHESES

Stage filters

Pivot the same store by Input · Thesis · Research · Debate · Control · Memory — or by agent (Trigger, Hypothesis, Bull, Bear, Guard, Curator).

Hypothesis cards

Each card surfaces regime tag, conviction, horizon, kill criteria count, the thesis statement, and the full reasoning paragraph — auditable forward-only.

Where did the P&L come from?

Attribution + live snapshot

Every order carries parent_hypothesis_id — P&L rolls up to the thesis

ATTRIBUTION DIMENSIONS

- › **By instrument** Top contributors / detractors
- › **By hypothesis** P&L rolls up to the thesis
- › **By regime quadrant** Reflation · Disinflation · Goldilocks · Stagflation
- › **By trigger source** Calendar · conviction · black swan · figure deviation · cron
- › **By approval outcome** Did rejections leave money on the table?
- › **By guard intervention** Did downsizing save money or cost it?

WHAT THE SYSTEM IS DOING NOW

Regime	e.g. Disinflation · $P(\text{growth}\uparrow)=0.32$
Risk-off	Probability + gate tier
Conviction ledger	Closest dimension to firing
Last 3 triggers	What fired them
Open positions	With parent hypothesis text
Approval queue	Or '0 — pipeline isn't blocked'

What's next

Honest gaps - roadmap

Where we know we're light, and where we're headed

ARCHITECTURE

- › Fully accumulator-based triggers
- › Earnings transcripts + central-bank statement ingestion
- › Cross-asset correlation cap as a hard rule

MODELS

- › Volatility regime as a 3rd nowcaster dimension
- › Embedding-based Polymarket contract matching
- › Backtest the technical readiness multiplier

OPERATIONS

- › Wire to real broker (Alpaca / IBKR)
- › Volatility-targeted position sizing
- › SSE streaming to dashboard

OPEN RESEARCH

Does 'buy fear at extremes' work historically? · Should hard rules be amendable by 2-of-N human approval?